Data Analytics & Data Engineering

Project: NAMO

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Introduction

This document provides an overview of the **Data Analytics & Data Engineering** process followed in the NAMO project. It includes the technologies used, the workflow, and key insights.

Tech Stack & Tools

1. **Mage AI**

- A powerful platform designed to build and deploy machine learning models efficiently.
- It allows data ingestion, transformation in an easy-to-use interface.

2. **AWS Redshift Table**

- A columnar storage database optimized for large-scale data processing.
- Ensures high performance, scalability, and efficiency in querying big data.

3. **Amazon S3 (Simple Storage Service)**

- A scalable storage service used for backup, archiving, and hosting large datasets.
- Stores logs, videos, images, and structured/unstructured data securely.

4. **Visualization Tool: Apache Superset**

- An open-source data visualization and business intelligence tool.
- Provides interactive dashboards and insights for data analysis.

5. **Programming Language: Python**

- Python is used for data processing, analysis, and automation.
- Supports multiple libraries like Pandas, NumPy, and SciPy for efficient computations.

Workflow

The following steps outline the Data Analytics and Engineering workflow:

1. **Understand the Problem:** Gather data from multiple sources such as databases, APIs, and logs.
2. **Data Cleaning & Pre-processing:** Remove inconsistencies, handle missing values, and format data for analysis.
3. **Data Analysis & Modelling**: Use analytical techniques to derive insights and build predictive models.
4. **Insight Generation & Interpretation:** Identify patterns and trends to support decision-making.
5. **Iteration & Improvement:** Optimize workflows based on feedback and new data.

Conclusion

The combination of **Mage AI, AWS Redshift, S3, Apache Superset, and Python** enables a **robust data analytics and engineering framework**. This ensures efficient data processing, visualization, and decision-making in the NAMO project.